ELSEVIER

Journal of Alloys and Compounds 297 (2000) 185–191

Journal of ALLOYS AND COMPOUNDS

www.elsevier.com/locate/jallcom

Thermodynamic modeling of the Pd–Y system

Z. Du*, H. Yang, C. Li

Department of Materials Science and Engineering, University of Science and Technology Beijing, Beijing 100083, P.R. China

Received 27 April 1999; accepted 7 September 1999

Abstract

The Pd–Y system was critically assessed using the CALPHAD technique. The solution phases (liquid, b.c.c., f.c.c. and h.c.p.) were modeled using the Redlich–Kister equation. The intermetallic compounds Pd_3Y and PdY, which have a homogeneity range, were treated as the phases MPd_3 and MY, respectively, by a two-sublattice model with Pd in MPd_3 and Y in MY on one sublattice and M on the other, where M is used as an abbreviation for a mixture of Pd and Y; the others as stoichiometric compounds. A set of self-consistent thermodynamic parameters of the Pd–Y system was obtained. © 2000 Elsevier Science S.A. All rights reserved.

Keywords: Pd–Y system; Thermodynamic properties; Thermodynamic model

1. Introduction

The intermetallic compounds formed by rare earth elements (RE) and transition metals are of particular interest regarding their potential usage as high value functional materials, such as permanent magnets and hydrogen storage materials [1,2]. The palladium-rich rare earth solid solution alloys are of very great interest because of their potential applications as hydrogen diffusion membranes for purification and isotope separation [3–14]. The Pd–Y system is very important from this standpoint. It is a major binary constituent of the Y–Pd–B–C quaternary system, in which rapidly quenched palladium-based borocarbides with large fractions of the superconducting phase have been obtained [15]. To understand the physical properties and the technological applications of these compounds, it is necessary to obtain a better knowledge of the thermodynamic properties of this technically relevant system.

This work deals with an assessment of the thermodynamic description of the Pd–Y system, which forms part of a general programme on the thermodynamic optimization of Pd–RE–H(D) systems, by means of the CALPHAD (CALculation of PHAse Diagram) technique. In this method, the thermodynamic properties of alloy systems are studied using thermodynamic models for the Gibbs energy of individual phases. The thermodynamic parameters involved in the models are optimized from the experimental thermodynamic and phase diagram information.

2. Thermodynamic models

2.1. Unary phases

The Gibbs energy function ${}^0G_i^{\phi}(T) = G_i^{\phi}(T) - H_i^{\mathrm{SER}}$ for the element i (i = Pd, Y) in the phase ϕ (ϕ = liquid, b.c.c., f.c.c. or h.c.p.) is described by an equation of the following form:

$$ {}^0G_i^{\phi}(T) = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \quad (1) $$

where H_i^{SER} is the molar enthalpy of element i at 298.15 K in its Standard Element Reference (SER) state, f.c.c. for Pd and h.c.p. for Y. The Gibbs energy of element i, ${}^0G_i^{\phi}(T)$, in its SER state is denoted by GHSER_i, i.e.

$$ \mathrm{GHSER}_{\mathrm{Pd}} = G_{\mathrm{Pd}}^{\mathrm{f.c.c.}}(T) - H_{\mathrm{Pd}}^{\mathrm{SER}} \quad (2) $$

$$ \mathrm{GHSER}_{\mathrm{Y}} = G_{\mathrm{Y}}^{\mathrm{h.c.p.}}(T) - H_{\mathrm{Y}}^{\mathrm{SER}} \quad (3) $$

In the present work, the Gibbs energy functions are taken from the SGTE compilation of Dinsdale [16] for Pd and Shen et al. [17] for Y.

*Corresponding author. Tel./fax: +86-10-62333772.
E-mail address: calphad@public.fhnet.cn.net (Z. Du)

0925-8388/00/$ – see front matter © 2000 Elsevier Science S.A. All rights reserved.
PII: S0925-8388(99)00582-4

2.2. Solution phases

In the Pd–Y system, there are four solution phases: liquid, b.c.c., f.c.c. and h.c.p. Their Gibbs energies are described by the following expression:

$$G_m^{\phi} - x_{\mathrm{Pd}}H_{\mathrm{Pd}}^{\mathrm{SER}} - x_{\mathrm{Y}}H_{\mathrm{Y}}^{\mathrm{SER}} = x_{\mathrm{Pd}}\,{}^0G_{\mathrm{Pd}}^{\phi}(T) + x_{\mathrm{Y}}\,{}^0G_{\mathrm{Y}}^{\phi}(T) + RT(x_{\mathrm{Pd}}\ln x_{\mathrm{Pd}} + x_{\mathrm{Y}}\ln x_{\mathrm{Y}}) + {}^{\mathrm{E}}G_m^{\phi} + {}^{\mathrm{mg}}G_m^{\phi} \quad (4)$$

where ${}^{\mathrm{E}}G_m^{\phi}$ is the excess Gibbs energy, expressed by the Redlich–Kister polynomial

$${}^{\mathrm{E}}G_m^{\phi} = x_{\mathrm{Pd}}x_{\mathrm{Y}}\sum_i {}^iL^{\phi}(x_{\mathrm{Pd}} - x_{\mathrm{Y}})^i \quad (5)$$

where ${}^iL^{\phi}$ is the interaction parameter between elements Pd and Y, which is to be evaluated in the present work. Its general form is

$$L^{\phi} = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} \quad (6)$$

but only the first one or two terms are used. ${}^{\mathrm{mg}}G_m^{\phi}$ is the magnetic contribution to the Gibbs energy in solution phases, and is described by

$${}^{\mathrm{mg}}G_m^{\phi} = RT\ln(\beta^{\phi} + 1)f(\tau^{\phi}) \quad (7)$$

where β^{ϕ} is a quantity related to the total magnetic entropy, which in most cases is set equal to the Bohr magnetic moment per mole of atoms; τ^{ϕ} is defined as T/T_{c}^{ϕ}, and T_{c}^{ϕ} is the critical temperature for magnetic ordering, i.e. the Curie temperature (T_{C}) for ferromagnetic ordering or the Neél temperature (T_{N}) for antiferromagnetic ordering. They are described by the following expressions:

$$T_{\mathrm{c}}^{\phi} = x_{\mathrm{Pd}}\,{}^0T_{\mathrm{c\,Pd}}^{\phi} + x_{\mathrm{Y}}\,{}^0T_{\mathrm{c\,Y}}^{\phi} + x_{\mathrm{Pd}}x_{\mathrm{Y}}L_{T_{\mathrm{c}}}^{\phi} \quad (8)$$

$$\beta^{\phi} = x_{\mathrm{Pd}}\,{}^0\beta_{\mathrm{Pd}}^{\phi} + x_{\mathrm{Y}}\,{}^0\beta_{\mathrm{Y}}^{\phi} + x_{\mathrm{Pd}}x_{\mathrm{Y}}L_{\beta}^{\phi} \quad (9)$$

where $L_{T_{\mathrm{c}}}^{\phi}$ and L_{β}^{ϕ} are the magnetic interaction parameters between elements Pd and Y, here set to zero due to lack of experimental data for the Pd–Y system. The $f(\tau)$ represent the polynomials obtained by Hillert and Jarl [18] based on the magnetic specific heat of iron, i.e. for $\tau < 1$:

$$f(\tau) = 1 - [79\tau^{-1}/(140p) + 474/497(1/p - 1) \times (\tau^3/6 + \tau^9/135 + \tau^{15}/600)]/A \quad (10)$$

for $\tau > 1$:

$$f(\tau) = -(\tau^{-5}/10 + \tau^{-15}/315 + \tau^{-25}/1500)/A \quad (11)$$

where $A = 519/1125 + 11\,692/15\,975(1/p - 1)$ and p depends on the structure, 0.4 for the b.c.c. structure and 0.28 for the others.

2.3. Stoichiometric intermetallic compounds

The intermetallic compounds Pd_7Y, Pd_2Y, Pd_3Y_2, Pd_4Y_3, Pd_2Y_3, Pd_2Y_5 and PdY_3 are treated as stoichiometric compounds. The Gibbs energy for per mole of formula unit Pd_mY_n is expressed as:

$${}^0G_m^{\mathrm{Pd}_m\mathrm{Y}_n} = G_m^{\mathrm{Pd}_m\mathrm{Y}_n} - mH_{\mathrm{Pd}}^{\mathrm{SER}} - nH_{\mathrm{Y}}^{\mathrm{SER}} = m\mathrm{GHSER}_{\mathrm{Pd}} + n\mathrm{GHSER}_{\mathrm{Y}} + \Delta G_{\mathrm{f}}^{\mathrm{Pd}_m\mathrm{Y}_n} + {}^{\mathrm{mg}}G_m^{\mathrm{Pd}_m\mathrm{Y}_n} \quad (12)$$

where $\Delta G_{\mathrm{f}}^{\mathrm{Pd}_m\mathrm{Y}_n}$ is the Gibbs energy of formation for per mole of formula unit Pd_mY_n. Owing to lack of experimental measurements, it is assumed that the Neumann–Kopp rule applies to the heat capacity, i.e. $\Delta C_{\mathrm{p}} = 0$. Thus, $\Delta G_{\mathrm{f}}^{\mathrm{Pd}_m\mathrm{Y}_n}$ can be given by the following expression:

$$\Delta G_{\mathrm{f}}^{\mathrm{Pd}_m\mathrm{Y}_n} = a + bT \quad (13)$$

where the parameters a and b were to be evaluated in the present work. The magnetic contribution to the Gibbs energy ${}^{\mathrm{mg}}G_m^{\mathrm{Pd}_m\mathrm{Y}_n}$ is included in the Gibbs energy ${}^0G_m^{\mathrm{Pd}_m\mathrm{Y}_n}$, because the Curie temperature of all compounds in the Pd–Y system is below 298.15 K.

2.4. Pd_3Y and PdY compounds

The intermetallic compounds Pd_3Y and PdY, which have a homogeneity range, are treated as the phases MPd_3 and MY, respectively, by a two-sublattice model [19,20] with Pd in MPd_3 and Y in MY on one sublattice and M on the other, where M is used as an abbreviation for a mixture of Pd and Y. The Gibbs energies per mole of formula unit MPd_3 and MY are given respectively by the following expressions:

$${}^0G_m^{\mathrm{MPd}_3} = G_m^{\mathrm{MPd}_3} - H_{\mathrm{MPd}_3}^{\mathrm{SER}} = y_{\mathrm{Pd}}\,{}^0G_{\mathrm{Pd:Pd}}^{\mathrm{MPd}_3} + y_{\mathrm{Y}}\,{}^0G_{\mathrm{Y:Pd}}^{\mathrm{MPd}_3} + RT(y_{\mathrm{Pd}}\ln y_{\mathrm{Pd}} + y_{\mathrm{Y}}\ln y_{\mathrm{Y}}) + y_{\mathrm{Pd}}y_{\mathrm{Y}}\sum_i {}^iL_{\mathrm{Pd,Y:Pd}}^{\mathrm{MPd}_3}(y_{\mathrm{Pd}} - y_{\mathrm{Y}})^i \quad (14)$$

$${}^0G_m^{\mathrm{MY}} = G_m^{\mathrm{MY}} - H_{\mathrm{MY}}^{\mathrm{SER}} = y_{\mathrm{Pd}}\,{}^0G_{\mathrm{Pd:Y}}^{\mathrm{MY}} + y_{\mathrm{Y}}\,{}^0G_{\mathrm{Y:Y}}^{\mathrm{MY}} + RT(y_{\mathrm{Pd}}\ln y_{\mathrm{Pd}} + y_{\mathrm{Y}}\ln y_{\mathrm{Y}}) + y_{\mathrm{Pd}}y_{\mathrm{Y}}\sum_i {}^iL_{\mathrm{Pd,Y:Y}}^{\mathrm{MY}}(y_{\mathrm{Pd}} - y_{\mathrm{Y}})^i \quad (15)$$

where $H_{\mathrm{MPd}_3}^{\mathrm{SER}}$ and $H_{\mathrm{MY}}^{\mathrm{SER}}$ are abbreviations for $(y_{\mathrm{Pd}} + 3)H_{\mathrm{Pd}}^{\mathrm{SER}} + y_{\mathrm{Y}}H_{\mathrm{Y}}^{\mathrm{SER}}$ and $y_{\mathrm{Pd}}H_{\mathrm{Pd}}^{\mathrm{SER}} + (y_{\mathrm{Y}} + 1)H_{\mathrm{Y}}^{\mathrm{SER}}$, and y_{Pd} and y_{Y} are the site fractions of Pd and Y on the M sublattice. The four parameters denoted ${}^0G_{*:*}^{*}$ (also called compound energies) are expressed relative to the Gibbs energies of pure f.c.c. Pd and h.c.p. Y in their SER state.

The ${}^{i}L$ represent the interaction parameters between elements Pd and Y on the M sublattice.

3. Experimental information

The complete phase diagram of the Pd–Y system was first measured by Loebich and Raub [21]. They found seven intermetallic compounds in the Pd–Y system and called them Pd_3Y, Pd_2Y, Pd_3Y_2, Pd_5Y_4, PdY, Pd_2Y_3 and Pd_2Y_5, where Pd_3Y, Pd_5Y_4, PdY and Pd_2Y_5 melted congruently and Pd_2Y, Pd_3Y_2 and Pd_2Y_3 formed by peritectic reactions. Later, Pd_5Y_4 was identified to be Pd_4Y_3 by Palenzona and Iandelli [22]. PdY_3 and Pd_7Y were found to exist by Sanjines-Zeballos et al. [23] and Takao et al. [24], respectively. Kuentzler and Loebich [25] presented a revised phase diagram and discussed the stability of the existing ordered structures.

The homogeneity range of the intermetallic compound Pd_3Y and the solubility of Y in f.c.c.(Pd) were measured by Loebich and Raub [21] and Harris and Norman [26]. No appreciable solid solubility of Pd in Y was observed [21]. Loebich and Raub [21] indicated the presence of polymorphic transformation and a small range of homogeneity for PdY (51–52 at.% Y). Considering, however, the subsequent work of Kuentzler and Loebich [25] and Palenzona and Cirafici [27], the homogeneity range of PdY should be 50–51 at.% Y and the maximum melting point attributed to the ideal 50 at.% Y composition [28].

The thermodynamic activities of Y in f.c.c.(Pd) and within the homogeneity range of the intermetallic compound Pd_3Y at 1023 K were determined by Paasch and Schaller [29], and partial molar excess Gibbs energies, excess Gibbs energies, excess entropies and the enthalpies of mixing were derived.

Recently, Alqasmi et al. [30] determined the thermodynamic activities of Y for whole compositions at 973 and 1073 K, the Gibbs energies at 1073 K and the enthalpies of formation for seven intermetallic compounds: Pd_3Y, Pd_2Y, Pd_3Y_2, Pd_4Y_3, PdY, Pd_2Y_3 and Pd_2Y_5.

4. Assessment procedure

Almost all of the above experimental information was selected for the evaluation of the thermodynamic model parameters.

The optimization was carried out by means of the Thermo-Calc software [31] which can handle various kinds of experimental data. The program works by minimizing an error sum where each of the selected data values is given a certain weight. The weight is chosen by personal judgment and changed by trial and error during the work until most of the selected experimental information is reproduced within the expected uncertainty limits.

The optimization was carried out in two steps. In the first treatment, Pd_3Y and PdY are assumed to be stoichiometric compounds; in the second treatment they are treated as MPd_3 and MY by a two-sublattice model which is described in Section 2.4. The parameters obtained from the first treatment were used as starting values for the second treatment.

No solubility is assumed in the b.c.c.(Y) and h.c.p.(Y) solutions, which is realized by assigning a large positive interaction parameter, i.e.

$$ {}^{0}L^{\text{b.c.c.}}_{\text{Pd,Y}} = {}^{0}L^{\text{h.c.p.}}_{\text{Pd,Y}} = 10^5 \text{ J mol}^{-1} \tag{16} $$

For f.c.c., the temperature dependence of the solubility of Y in f.c.c.(Pd) is well established [21,26]. The coefficients a_0 and b_0 can be reliably obtained from the experimental data.

For the liquid, at least a_0 and b_0 in Eq. (4) can be adjusted because the liquidus was measured accurately over the whole composition range. In the present work, it was found that a_1 and b_1 should also be introduced in order to described the properties of the liquid satisfactory.

The quantities ${}^{0}G^{\text{MPd}_3}_{\text{Pd:Pd}}$ in Eq. (14) and ${}^{0}G^{\text{MY}}_{\text{Y:Y}}$ in Eq. (15) describe a metastable form of pure Pd with the Pd_3Y structure and pure Y with the PdY structure, respectively. The parameters are optimized to be a positive a_0 above the GHSER_i (i = Pd or Y). But for ${}^{0}G^{\text{MY}}_{\text{Y:Y}}$, the parameters a_0 and b_0 have to be optimized because the polymorphic transformation of the intermetallic compounds PdY is determined [21].

Table 1
Thermodynamic parameters of the Pd–Y system[a]

Liquid	${}^{0}L^{\text{liq.}} = -351\,248 + 31.479T$
	${}^{1}L^{\text{liq.}} = -261\,969 + 97.507T$
b.c.c.	${}^{0}L^{\text{b.c.c.}} = 100\,000$
f.c.c.	${}^{0}G^{\text{f.c.c.}}_{\text{Y}} - \text{GHSER}_{\text{Y}} = +199\,766$
	${}^{0}L^{\text{f.c.c.}} = -699\,921 + 70.141T$
h.c.p.	${}^{0}L^{\text{h.c.p.}} = 100\,000$
Pd_7Y	$\Delta G^{\text{Pd}_7\text{Y}}_{\text{f}} = -465\,505 + 103.545T$
MPd_3	${}^{0}G^{\text{MPd}_3}_{\text{Pd:Pd}} - 4\text{GHSER}_{\text{Pd}} = +10\,492$
	${}^{0}G^{\text{MPd}_3}_{\text{Y:Pd}} - 3\text{GHSER}_{\text{Pd}} - \text{GHSER}_{\text{Y}} = -384\,317 + 50.046T$
	${}^{0}L^{\text{MPd}_3}_{\text{Pd,Y:Pd}} = -51\,789$
Pd_2Y	$\Delta G^{\text{Pd}_2\text{Y}}_{\text{f}} = -283\,002 + 24.568T$
αPd_3Y_2	$\Delta G^{\alpha\text{Pd}_3\text{Y}_2}_{\text{f}} = -464\,736 + 27.610T$
βPd_3Y_2	$\Delta G^{\beta\text{Pd}_3\text{Y}_2}_{\text{f}} = -461\,862 + 25.429T$
Pd_4Y_3	$\Delta G^{\text{Pd}_4\text{Y}_3}_{\text{f}} = -646\,108 + 30.777T$
αMY	${}^{0}G^{\alpha\text{MY}}_{\text{Pd:Y}} - \text{GHSER}_{\text{Pd}} - \text{GHSER}_{\text{Y}} = -181\,437 + 8.527T$
	${}^{0}G^{\alpha\text{MY}}_{\text{Y:Y}} - 2\text{GHSER}_{\text{Y}} = +306\,191$
	${}^{0}L^{\alpha\text{MY}}_{\text{Pd,Y:Y}} = -277\,205$
βMY	${}^{0}G^{\beta\text{MY}}_{\text{Pd:Y}} - \text{GHSER}_{\text{Pd}} - \text{GHSER}_{\text{Y}} = -181\,166 + 8.290T$
	${}^{0}G^{\beta\text{MY}}_{\text{Y:Y}} - 2\text{GHSER}_{\text{Y}} = +366\,767 - 15.453T$
	${}^{0}L^{\beta\text{MY}}_{\text{Pd,Y:Y}} = -315\,275$
Pd_2Y_3	$\Delta G^{\text{Pd}_2\text{Y}_3}_{\text{f}} = -439\,067 + 68.860T$
Pd_2Y_5	$\Delta G^{\text{Pd}_2\text{Y}_5}_{\text{f}} = -479\,842 + 95.807T$
PdY_3	$\Delta G^{\text{PdY}_3}_{\text{f}} = -249\,018 + 55.580T$

[a] In J/(mol of formula units).

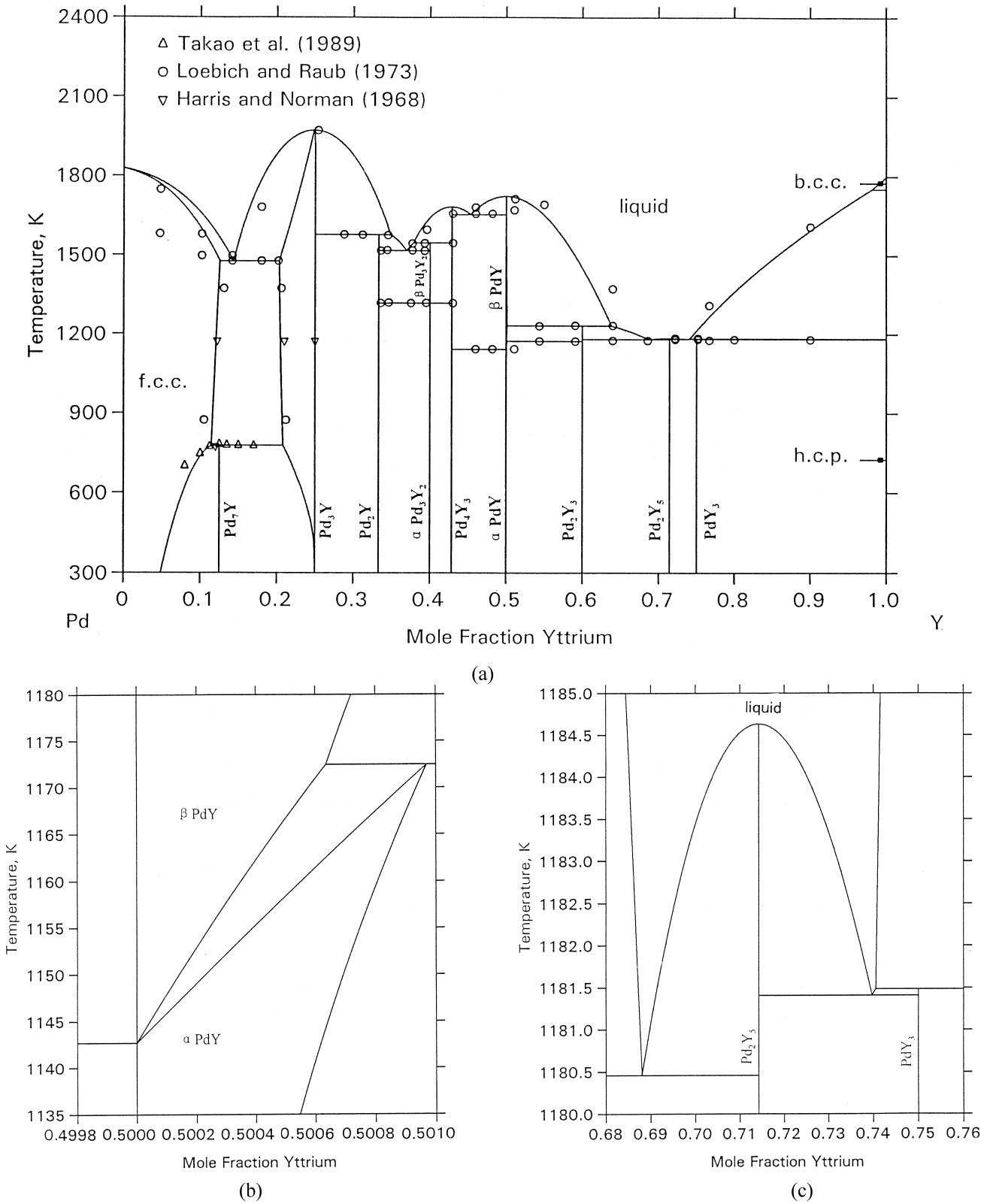

Fig. 1. (a) The Pd–Y phase diagram as calculated in the present work with experimental data [21,24,26]. (b) Enlarged section of (a). (c) Enlarged section of (a).

Table 2
Invariant reactions in the Pd–Y system

Reaction	Present work				Ref. [21]			
	T (K)	$x(Y)$			T (K)	$x(Y)$		
Liq.+b.c.c.(Y)→h.c.p.(Y)	1752	0.9808	1.0000	1.0000	1751	0.98	1.00	1.00
Liq.+h.c.p.(Y)→PdY_3[a]	1181	0.7405	1.0000	–	1181	0.745	1.00	–
Liq.→Pd_2Y_5+PdY_3[a]	1181	0.7397	–	–	1168	0.73	–	–
Liq.→Pd_2Y_5	1185	–	–		1188	–	–	
Liq.→Pd_2Y_5+Pd_2Y_3	1180	0.6880	–	–	1176	0.70	–	–
Liq.+βPdY→Pd_2Y_3	1232	0.6384	0.5016	–	1233	0.66	0.52	–
Liq.→βPdY	1724	0.4985	–		1723	0.50	–	
Liq.→βPdY+Pd_4Y_3	1655	0.4535	0.5000	–	1658	0.47	0.51	–
Liq.→Pd_4Y_3	1684	–	–		1683	–	–	
Liq.+Pd_4Y_3→βPd_3Y_2	1548	0.3782	–	–	1547	0.38	–	–
Liq.→βPd_3Y_2+Pd_2Y	1518	0.3692	–	–	1518	0.37	–	–
Liq.+Pd_3Y→Pd_2Y	1579	0.3484	0.2500	–	1578	0.34	0.25	–
Liq.→Pd_3Y	1973	0.2490	–		1973	0.25	–	
Liq.→Pd_3Y+f.c.c.(Pd)	1478	0.1436	0.2031	0.1257	1478	0.15	0.205	0.13
F.c.c.(Pd)+Pd_3Y→Pd_7Y[b]	777	0.1145	0.2082	–	777	0.105	0.21	–
αPd_3Y_2→βPd_3Y_2	1318	–	–		1318	–	–	
βPdY+Pd_2Y_3→αPdY	1173	0.5006	–	0.5010	1173	0.52	–	0.52
βPdY→Pd_4Y_3+αPdY	1143	0.5000	–	0.5000	1143	0.51	–	0.51

[a] Experimental data taken from Ref. [25].
[b] Experimental data taken from Ref. [24].

5. Results and discussions

A thermodynamic description of the Pd–Y system obtained in the present work is shown in Table 1. The Pd–Y phase diagram calculated by means of the thermodynamic parameters is presented in Fig. 1a with experimental data [21,24,26]. Figs. 1b,c are enlarged sections of Fig. 1a. The calculations agree well with the experimental results.

The invariant equilibria in the Pd–Y system are listed in Table 2. As shown in the table, satisfactory agreement is obtained between the calculations and experiment, where the largest uncertainty is about 4°C in the invariant temperature of the reaction liq.→Pd_2Y_5+Pd_2Y_3. In the reaction liq.→Pd_2Y_5+PdY_3, the uncertainty in the invariant temperature is about 13°C, which, however, was estimated by Kuentzler and Loebich [25]. In view of the

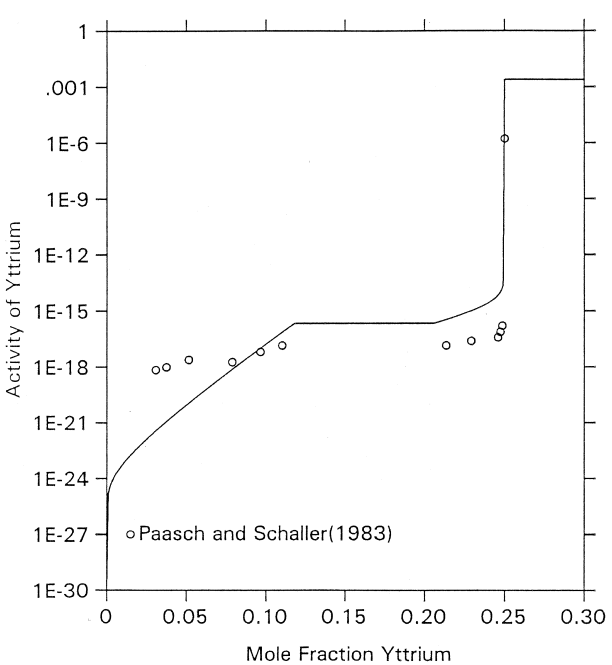

Fig. 2. Calculated activity of Y at 1023 K in the Pd–Y system with experimental data [29].

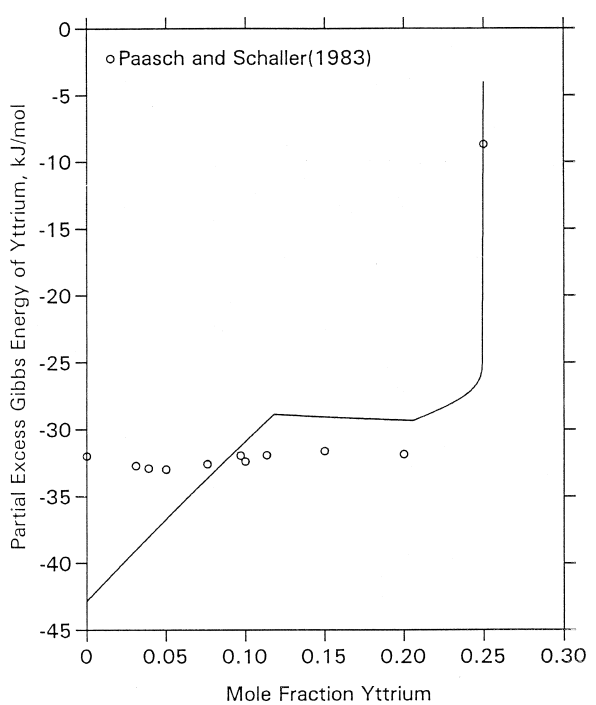

Fig. 3. Calculated partial molar excess Gibbs energy of Y at 1023 K in the Pd–Y system with experimental data [29].

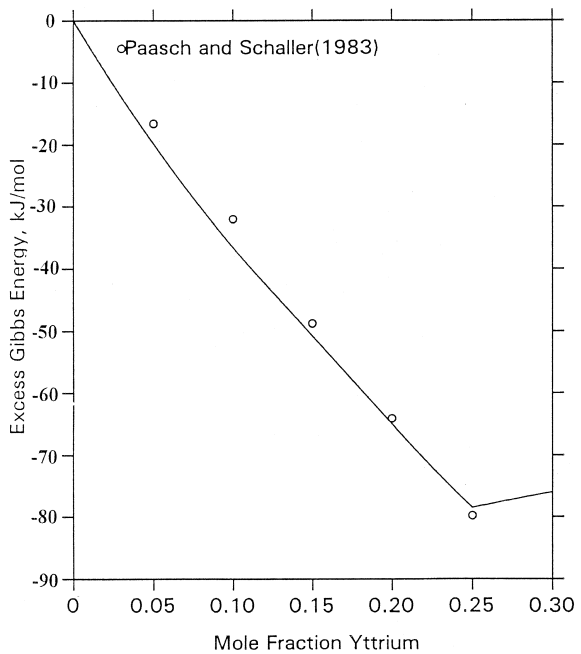

Fig. 4. Calculated excess Gibbs energy at 1023 K in the Pd–Y system with experimental data [29].

estimated experimental errors (about 1–2 at.%), all experimental invariant reaction compositions in the Pd–Y system are well reproduced.

Figs. 2–5 show the calculated thermodynamic activities and partial molar excess Gibbs energies of Y, excess Gibbs energies and excess entropies in f.c.c.(Pd) and Pd_3Y at 1023 K with the experimental data [29]. Fig. 6a,b present

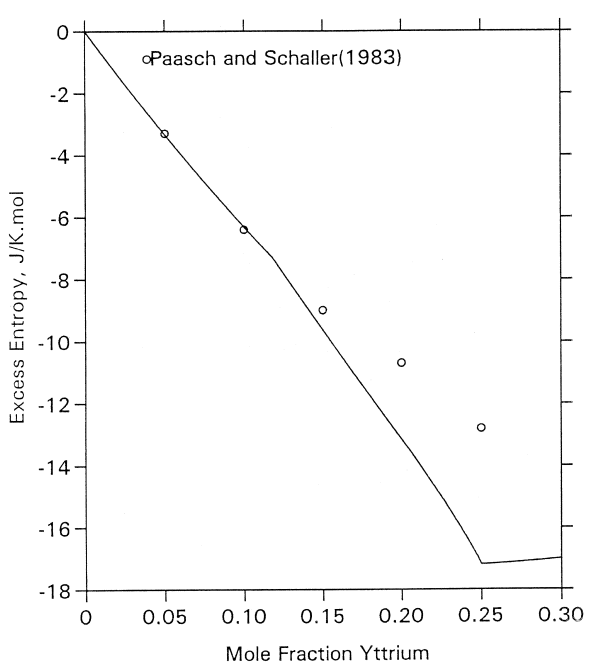

Fig. 5. Calculated excess entropy at 1023 K in the Pd–Y system with experimental data [29].

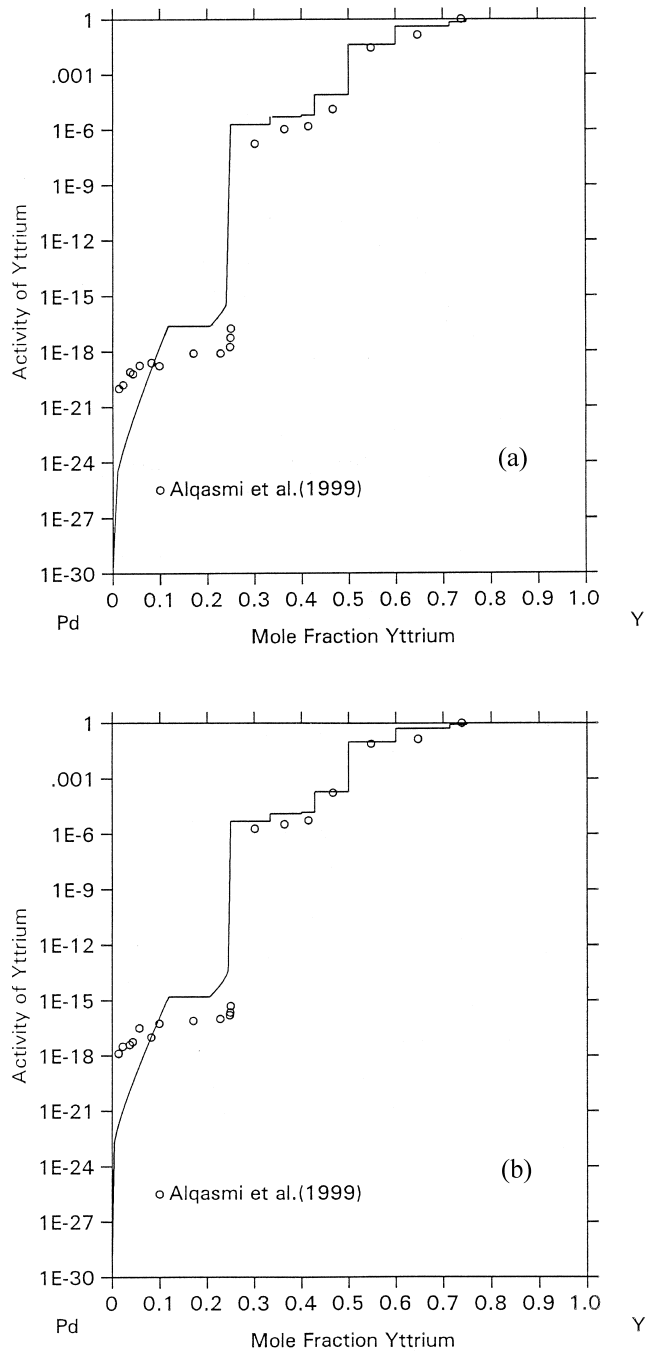

Fig. 6. (a) Calculated activity of Y at 973 K in the Pd–Y system with experimental data [30]. (b) Calculated activity of Y at 1073 K in the Pd–Y system with experimental data [30].

the calculated thermodynamic activities of Y at 973 and 1073 K with the experimental data [30]. Fig. 7 shows the enthalpies of formation in the Pd–Y system with the experimental data [29,30]. Fig. 8 gives the Gibbs energies at 1073 K with the experimental data [30]. Within experimental uncertainties, as shown in Figs. 2–8, all experimental data are very well reproduced by the present calculation.

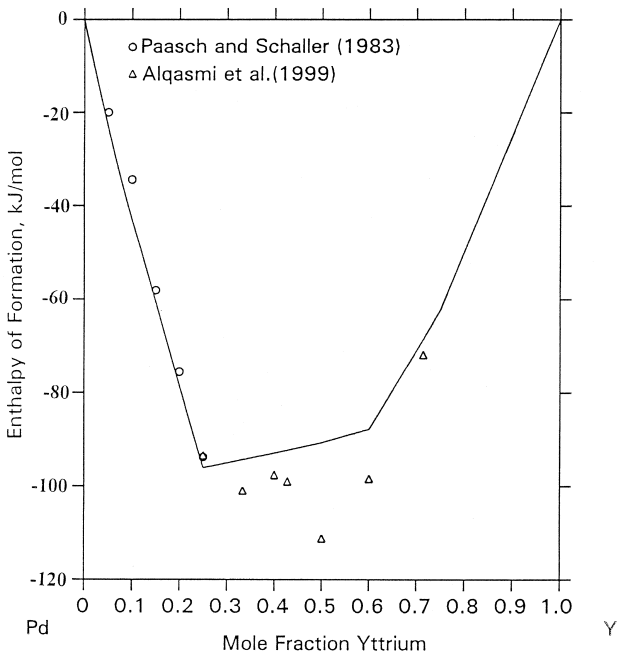

Fig. 7. Calculated enthalpy of formation in the Pd–Y system with experimental data [29,30].

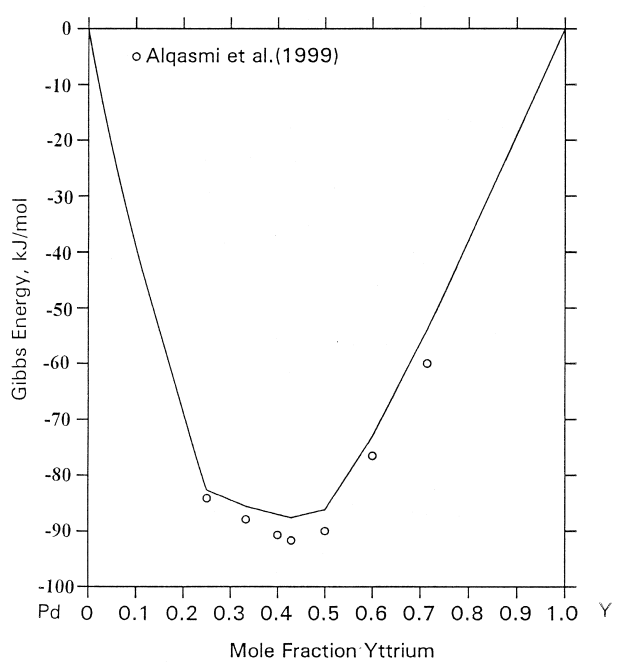

Fig. 8. Calculated Gibbs energy at 973 K in the Pd–Y system with experimental data [30].

6. Conclusions

The phase relations and thermodynamic description of the Pd–Y system were evaluated using experimental information available in the literature. A consistent set of thermodynamic parameters was derived. With the available thermodynamic description, one can now make various calculations of practical interest.

Acknowledgements

The authors would like to express their gratitude to Thermo-Calc AB for supplying the Thermo-Calc software. This work was supported by the National Natural Science Foundation of China (NSFC) (grant No. 59871008).

References

[1] K.H.J. Buschow, Rep. Prog. Phys. 40 (1977) 1179.
[2] K.H.J. Buschow, Mater. Res. Bull. 19 (1984) 935.
[3] M. Rorman, I.R. Harris, J. Less-Common Met. 18 (1969) 333.
[4] Y. Sakamoto, K. Kajihara, Y. Fukusaka, T.B. Flanagan, Z. Phys. Chem. Neue Folge, Bd. 159 (1988) S61.
[5] M. Ege, H. Kronmuller, J. Alloys Comp. 261 (1997) 231.
[6] S. Ramaprabhu, N. Rajalakshmi, A. Weiss, Bur. Bunsenges. Phys. Chem. 93 (1989) 686.
[7] F.L. Chen, M. Furukawa, Y. Sakamoto, J. Less-Common Met. 155 (1989) 173.
[8] O. Yoshinari, H. Matsuda, K. Fukuhara, K. Tanaka, Mater. Trans. JIM 38 (1997) 508.
[9] M. Ege, G. Hauptmann, W. Ulfert, H. Kronmuller, J. Alloys Comp. 211/212 (1994) 222.
[10] K.J. Bryden, J.Y. Ying, Acta Mater. 44 (1996) 3847.
[11] P.A. Poyser, M. Kemali, D.K. Ross, J. Alloys Comp. 253/254 (1997) 175.
[12] M. Yoshihara, R.B. McLellan, J. Phys. Chem. Solids 49 (1988) 401.
[13] R.B. McLellan, Phys. Status Solidi (a) 116 (1989) K157.
[14] S. Ramaprabhu, R. Leiberich, A. Weiss, Z. Phys. Chem. Neue Folge, Bd. 161 (1989) S83.
[15] V. Strom, K.S. Kim, A.M. Grishin, K.V. Rao, J. Appl. Phys. 79 (1996) 5860.
[16] A.T. Dinsdale, CALPHAD 15 (1991) 317.
[17] H. Shen, W. Zhang, R. Wang, Q. Liu, Z. Du, Rare Met. 14 (1995) 193.
[18] M. Hillert, M. Jarl, CALPHAD 2 (1978) 227.
[19] M. Hillert, L.I. Staffansson, Acta Chem. Scand. 24 (1970) 3618.
[20] B. Sundman, J. Agren, J. Phys. Chem. Solids 42 (1981) 297.
[21] Q. Loebich Jr., E. Raub, J. Less-Common Met. 30 (1973) 47.
[22] A. Palenzona, A. Iandelli, J. Less-Common Met. 34 (1974) 121.
[23] R. Sanjines-Zeballos, B. Chabot, E. Parthe, J. Less-Common Met. 72 (1980) 17.
[24] K. Takao, Y. Sakamoto, M. Yoshida, J. Less-Common Met. 152 (1989) 115.
[25] R. Kuentzler, O. Loebich, J. Less-Common Met. 106 (1985) 335.
[26] I.R. Harris, M. Norman, J. Less-Common Met. 15 (1968) 285.
[27] A. Palenzona, S. Cirafici, Thermochim. Acta 12 (1975) 267.
[28] G. Borzone, G. Cacciamani, R. Ferro, CALPHAD 14 (1990) 139.
[29] S. Paasch, H.-J. Schaller, Ber. Bunsenges. Phys. Chem. 87 (1983) 812.
[30] R.A. Alqasmi, S. Paasch, H.J. Schaller, J. Alloys Comp. 283 (1999) 173.
[31] B. Sundman, B. Jansson, J.-O. Andersson, CALPHAD 9 (1985) 153.